DATE 18/10/26 OS

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How OS Boots up?

(i) Power on computer:

→ Power supply goes to all the external devices connected,
Ex) Monitor, CPU, etc.

(ii) CPU loads BIOS or UEFI:

→ BIOS is older version, UEFI is Newer version.

→ These both are firmware interfaces that help computer get started.

→ Performs tasks such as:
- Init. essential hardware
- Perform hardware checks
- Detect available storage devices

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(iii) BIOS/UEFI hand off to Boot device.

Next, BIOS/UEFI searches for the Bootloader in Bootdevices

Bootdevices include → Disk (HDD/SSD)
→ CD
→ USB Device

Bootloader → Loads the OS kernel into the Memory.

BIOS Searches for Bootloader in Master Boot Record (MBR) located in 0th index of Disk

UEFI searches for Bootloader in EFI partition

↓

Bootloader is present in a seperate partition called EFI partition.

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(iv) Bootloader loads the full OS:

Bootloader is the program that differs from OS to OS, it will be the one that initializes the OS & performs any important function.

//

Bootloader loads the kernel in the RAM.

↓ (control transferred to kernel)

Now, kernel is the executor & it initiates important features like:

- Memory Management
- Interrupt handle
- Device Drivers
- Process Management

Bootloaders of different OS:

- Mac → boot.efi
- Windows → bootloader.exe
- Linux → GRUB.